Question:

An automobile battery produces a potential difference (or "voltage") of 12 volts between its terminals. (It really consists of six 2 volt batteries following one after the other.) A headlightbulb is to be connected directly across the terminals of the battery and dissipate 40 watts of joule heat. What current will it draw and what must its resistance be?

Answer Choices:

A. 3.0 amps, 4 ohms

B. 2.8 amps, 3.2 ohms

C. 3.75 amps, 3.2 ohms

D. 4 amps, 3 ohms

E. 3.33 amps, 3.6 ohms

F. 5 amps, 2.4 ohms

G. 4.5 amps, 2.67 ohms

H. 2.2 amps, 5.45 ohms

I. 2.5 amps, 4.8 ohms

J. 3.5 amps, 4 ohms

Reasoning + Answer:

<think>

To find the current drawn by the headlight bulb, we'll use the formula P = IV, where P is the power dissipation, I is the current, and V is the voltage. Given that the power dissipation (P) is 40 watts and the voltage (V) is 12 volts, we can rearrange the formula to solve for I: I = P/V. Substituting the given values, we get I = 40/12. This simplifies to I = 3.33 amps. Once we have the current, we can use Ohm's law to find the resistance (R): R = V/I. Substituting the known values, we get R = 12/3.33. This simplifies to R = 3.6 ohms. Therefore, the current drawn by the headlight bulb is 3.33 amps, and its resistance must be 3.6 ohms. So, the answer choice that matches these values is E. 3.33 amps, 3.6 ohms.</think><answer>E</answer>